Memory-Driven Contrastive Embedding Enhancement for Fine-Grained Open-Set Semi-Supervised Learning

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Abstract

Existing VLM-based open-set semi-supervised learning (OSSL) methods primarily rely on coarse-grained class-level semantics, thereby hampering the accurate identification of in-distribution (ID) versus out-of-distribution (OOD) samples. To overcome this issue, we propose a novel memory-driven contrastive embedding enhancement approach by integrating fine-grained visual contrastive cues into class-level textual side. Concretely, we design a memory bank to construct and maintain a diverse collection of the most distinctive fine-grained visual embeddings. Guided by the memory bank, visual information is incorporated into the textual side through cross-modal contrastive learning, enabling more discriminative fine-grained separation between ID and OOD samples. Consequently, the learned OSSL model exhibits improved contrastive discriminability across known and unknown categories. Extensive experiments demonstrate that our method achieves state-of-the-art (SOTA) performance on multiple fine-grained datasets. The code is available at https://github.com/njustkmg/NeurIPS26-MDCEE.

1 Introduction

Semi-supervised learning (SSL) [10] is a prominent learning paradigm, as it effectively exploits both labeled and unlabeled data. However, real-world settings are often open-world, rendering the standard SSL assumption of identical training and test distributions invalid. This indicates that unlabeled data in real-world scenarios frequently include a substantial number of previously unseen classes. This problem setting, termed open-set semi-supervised learning (OSSL) [34, 35, 24, 16, 1], has gained increasing research interest in recent years.

To address the challenges of OSSL, extensive efforts have been made in the literature. Early approaches [2, 20, 29] rely on traditional deep learning models, which provide limited semantic priors and are therefore less reliable under open-set noise. Recent advances in foundation models have brought new perspectives to OSSL. Large-model-based approaches [16] enhance the ability to distinguish between in-distribution (ID) and out-of-distribution (OOD) samples by exploiting the rich semantic representations of foundation models such as CLIP [23]. Consequently, they have demonstrated strong effectiveness in OSSL applications.

Despite achieving strong performance, existing approaches [16] usually rely on coarse-grained class-level text representations, limiting their ability to capture fine-grained semantic differences. This issue becomes especially critical in open-world settings, as some OOD categories exhibit only fine-grained differences from ID categories. For example, distinguishing between a *Corgi*, a *Husky*, and a *German Shepherd* cannot be reliably achieved using class-level semantics alone. By incorporating fine-grained attributes such as a *Corgi's short legs* and *golden fur*, the discrimination task becomes substantially

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easier, as illustrated in Figure 1. This raises a natural question of how to obtain fine-grained and distinctive representations. Such subtle characteristics are inherently encoded in visual images. Accordingly, a natural solution is to extract fine-grained visual features and leverage them to enhance text-side representation learning, thereby injecting sufficiently discriminative information into textual representations.

Motivated by this insight, in this paper, we introduce a memory-driven contrastive embedding enhancement method that injects fine-grained visual contrastive cues into class-level text representations. First, to preserve fine-grained visual information, we design a class-level memory bank. Each memory vector is updated by incorporating patch-level visual features that are most relevant to the corresponding text cues. As a result, each memory vector aggregates the most discriminative visual cues of its associated class. Then, leveraging the fine-grained visual memory bank, we enhance in-class textual embeddings by incorporating discriminative visual cues, thereby enriching class semantics and improving the model's ability to distinguish subtle differences. Meanwhile, we combine dissimilar visual cues from the memory bank with base textual embeddings to construct a poisoned out-of-class text feature, further strengthening discriminative capability. By constructing these two contrastive text features, we inject rich and fine-grained visual information into textual embeddings, which improves the model's ability to distinguish between in-distribution and out-of-distribution samples during training. Consequently, the proposed approach significantly improves discriminative performance in OSSL settings. Our main contributions are listed as follows:

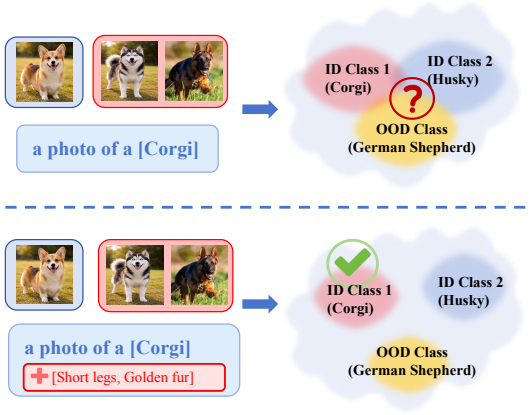


Figure 1: Issue of coarse-grained class-level textual cues. Top: cues with only class names fail to distinguish fine-grained classes and are vulnerable to OOD noise. Bottom: enriched cues with visual attributes (e.g., "short legs, golden fur"), achieving clearer decision boundaries and improved robustness against open-set noise.

- A novel fine-grained visual memory bank is designed, which captures discriminative visual cues that are closely associated with class-level semantics.
- A cross-modal contrastive learning mechanism that supports both in-class and out-of-class learning is proposed. By leveraging a fine-grained visual memory bank, rich visual cues are injected into textual embeddings, thereby improving the model's ability to distinguish between ID and OOD classes.
- Extensive experiments reveal that our proposed method achieves the best performance on various benchmarks compared with SOTA baselines.

2 Related Work

2.1 Open-Set Semi-Supervised Learning

Open-set semi-supervised learning (OSSL) targets a more realistic scenario, where unlabeled datasets inevitably contain OOD samples belonging to unknown categories absent from the labeled set [33, 32, 8, 7]. Early methods for OSSL, such as DS^{3}L [6], proposes a bi-level optimization framework that selectively utilizes unlabeled data by learning instance-wise weights. However, these early approaches suffer from notable limitations, particularly in handling uncertainty. In order to address these shortcomings, recent advancements have introduced increasingly sophisticated techniques. IOMatch [18] employs a multi-binary classifier with a closed-set classifier to produce unified open-set targets, treating all outliers explicitly as one class. The literature [37] proposes Adaptive Negative Evidential Deep Learning (ANEDL), utilizing evidential deep learning to quantify different types of uncertainties in predictions and adaptively optimize negative sampling strategies to differentiate inliers from outliers. CaliMatch [1] proposes an adaptive calibration framework that jointly calibrates the classifier and the OOD detector via adaptive label smoothing and temperature scaling. Recently,

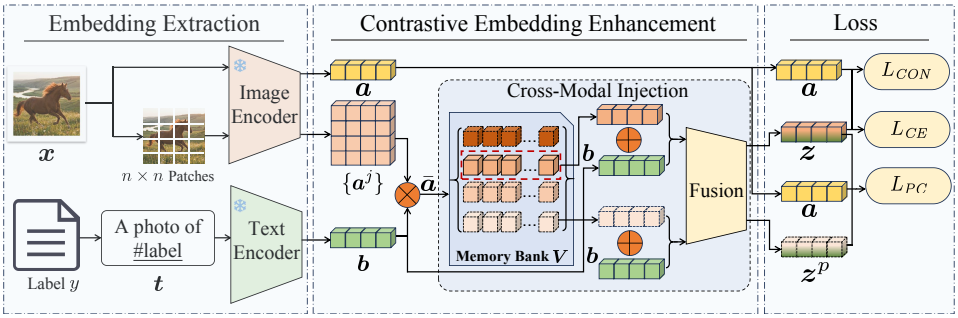


Figure 2: The architecture of our method. We design a **memory bank module** to maintain the most discriminative fine-grained visual cues. Based on this, a **contrastive learning mechanism** is designed to learn embeddings from both in-class and out-of-class perspectives.

CFSG-CLIP [16] introduces VLMs to OSSL. Despite these advancements, existing VLM-based OSSL methods [16] primarily rely on the inherent cross-modal alignment of pre-trained VLMs, where textual embeddings capture only coarse-grained semantic information. This coarse-grained representation is insufficient for distinguishing ID from near-OOD samples in fine-grained scenarios, where subtle visual differences are critical for accurate classification and OOD detection.

2.2 Pre-trained Vision-Language Models

Pre-trained vision-language models (VLMs) learn aligned image-text representations from large-scale image-text pairs, enabling open-vocabulary recognition by similarity matching in a shared embedding space. Representative dual-encoder contrastive pre-training paradigms include CLIP [23] and its scalable variants such as ALIGN [9], while more fine-grained alignment has also been explored by strengthening cross-modal modeling and capturing token-level correspondence [15, 36]. Built upon these backbones, downstream research mainly follows several recurring directions. One line focuses on efficient adaptation through minimal parameter updates or lightweight modules while preserving open-vocabulary capability [11, 39, 30, 5]. Another line extends VLMs from image-level to region-level recognition, learning region-aware representations to detect and ground novel categories beyond the training vocabulary [28, 17]. Despite these advances, existing CLIP-based methods primarily target standard supervised or few-shot scenarios and overlook the unique challenges of OSSL, where unlabeled data contains unknown categories and fine-grained ID-OOD discrimination is critical.

3 Method

In this section, we illustrate our proposed method in detail. The architecture of our method is shown in Figure 2.

3.1 Preliminary

Assume that we have a labeled dataset $\boldsymbol{X}_{\text{labeled}} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^{N}$, where $\boldsymbol{x}_i$ denotes the data sample drawn from the distribution $\mathcal{D}_{\text{labeled}}$, and $y_i \in \{1, \cdots, C\}$ indicates the class label with C classes. Furthermore, a large unlabeled dataset $\boldsymbol{X}_{\text{unlabeled}} = \{\boldsymbol{x}_i\}_{i=1}^{M}$ collected from the open-set environment is also available, where sample $\boldsymbol{x}_i$ is drawn from the distribution $\mathcal{D}_{\text{unlabeled}}$, which may contain both ID and OOD data. OSSL aims to correctly classify ID samples while effectively rejecting OOD samples using both the labeled set $\boldsymbol{X}_{\text{labeled}}$ and the unlabeled set $\boldsymbol{X}_{\text{unlabeled}}$.

In recent years, the rapid progress of large-scale pretrained VLMs has substantially advanced OSSL. Most existing OSSL methods [16] leverage the vision and text encoders of pretrained VLMs to obtain modality-specific embeddings. This process can be formulated as $\boldsymbol{a}_i = \phi^v(\boldsymbol{x}_i)$ and $\boldsymbol{b}_i = \phi^t(\boldsymbol{t}_i)$, where $\phi^v(\cdot)$ and $\phi^t(\cdot)$ respectively denote the vision and text encoders, and $\boldsymbol{t}_i$ represents a label-based textual prompt constructed from the class label y_i, e.g., "*A photo of* y_i". Based on the VLM-derived embeddings, the model performs ID-OOD discrimination by comparing samples against class-level textual prototypes.

Under this architecture, the text modality primarily captures coarse-grained visual semantics through cross-modal alignment, which limits its ability to reliably distinguish ID and OOD samples, especially in fine-grained open-set scenarios.

3.2 Visual Memory Bank

To enhance the discriminative capacity of the text modality for ID-OOD separation, we propose a memory bank-driven visual enhancement mechanism.

Given an input image $\boldsymbol{x}_i$, we partition it into n^2 non-overlapping patches, which are subsequently embedded as a sequence of patch-level tokens:

$$\forall j \in \{1, \cdots, n^2\},\ \boldsymbol{a}_i^j = \phi^v(\boldsymbol{x}_i^j) \in \mathbb{R}^d, \tag{1}$$

where $\boldsymbol{x}_i^j$ indicates the j-th patch, and d denotes the feature dimension. Furthermore, the textual embedding $\boldsymbol{b}_i$ is obtained by encoding the label-based textual prompt $\boldsymbol{t}_i$ constructed from y_i.

Based on the patch-level tokens and the textual embedding, we maintain a memory bank $\boldsymbol{V} \in \mathbb{R}^{C \times d}$, with one d-dimensional memory vector for each ID class. We use patch-text similarity as a class-relevance prior. Specifically, we first calculate the similarity score between $\{\boldsymbol{a}_i^j\}_{j=1}^{n^2}$ and $\boldsymbol{b}_i$:

$$\forall j \in \{1, \cdots, n^2\},\ s_i^j = \cos(\boldsymbol{a}_i^j, \boldsymbol{b}_i). \tag{2}$$

Based on the similarity scores, we select the top-K patch-level tokens and aggregate them into an averaged correlated memory slot by an alignment module function $g(\cdot)$:

$$\bar{\boldsymbol{a}}_i = \frac{1}{K} \sum_{k=1}^{K} g(\boldsymbol{a}_i^{r_k}; \theta_g), \tag{3}$$

where $r_k \in \Gamma_K(\{s_i^j\}_{j=1}^{n^2})$, and $\Gamma_K(\{s_i^j\}_{j=1}^{n^2})$ denotes the index set of the top-K patch-level tokens ranked by similarity scores, θ_g denotes the parameters of $g(\cdot)$.

During the training process, the normalized memory slot $\bar{\boldsymbol{a}}_i$ is used to update its corresponding class-wise memory bank $\boldsymbol{v}_{y_i} \in \boldsymbol{V}$ by exponential moving average (EMA) strategy at t-th iteration:

$$\boldsymbol{v}_{y_i}(t+1) = \begin{cases} \bar{\boldsymbol{a}}_i, & \text{if } t = 1, \\ \mu \boldsymbol{v}_{y_i}(t) + (1-\mu) \cdot \dfrac{\bar{\boldsymbol{a}}_i}{\|\bar{\boldsymbol{a}}_i\|}, & \text{otherwise.} \end{cases}$$

By doing so, the memory bank aggregates the most relevant fine-grained visual information, enabling stable and discriminative text-side enhancement.

3.3 Contrastive Embedding Enhancement

We next investigate how fine-grained visual cues can be integrated into the text modality to enhance ID-OOD discrimination. For coarse-grained textual cues, we incorporate fine-grained visual features from the same class to enrich semantic specificity, while simultaneously contrasting them with visually dissimilar features from other classes to promote class separability in the joint embedding space. Accordingly, we design a cross-modal injection strategy that jointly accounts for both in-class relevance and out-of-class discrepancy.

Specifically, we first utilize $\boldsymbol{v}_{y_i}$ to enrich the base textual embedding $\boldsymbol{b}_i$:

$$\boldsymbol{z}_i = \alpha h\left([\boldsymbol{v}_{y_i}; \boldsymbol{b}_i]; \theta_h\right) + \boldsymbol{b}_i, \tag{4}$$

where $[\cdot;\cdot]$ indicates the vector concatenation, $h(\cdot)$ denotes a fusion module function, θ_h denotes the parameters of $h(\cdot)$, and α is a weighting coefficient.

Simultaneously, we inject fine-grained visual cues from different classes into the base textual cue. Specifically, we randomly sample a memory vector from the memory bank corresponding to a class other than y_i:

$$\xi_i \leftarrow \{\boldsymbol{v}_1, \cdots, \boldsymbol{v}_C\} \setminus \{\boldsymbol{v}_{y_i}\}. \tag{5}$$

Here we use the notation "$\leftarrow$" to indicate that an element is randomly chosen from a given set. Then the textual embedding $\boldsymbol{b}_i$ is poisoned by:

$$\boldsymbol{z}_i^p = \alpha h\left([\xi_i; \boldsymbol{b}_i]; \theta_h\right) + \boldsymbol{b}_i. \tag{6}$$

By injecting fine-grained visual cues from both similar and dissimilar classes into the textual side, we enable a more robust contrastive learning mechanism. We first calculate the probability given $(\boldsymbol{x}_i, y_i)$ based on the enriched textual embedding $\boldsymbol{z}_i$:

$$p(y_i \mid \boldsymbol{x}_i) = \frac{\exp\left(\langle \boldsymbol{z}_i, \boldsymbol{a}_i \rangle / \tau\right)}{\sum_{j=1}^{C} \exp\left(\langle \boldsymbol{z}_j, \boldsymbol{a}_i \rangle / \tau\right)}, \tag{7}$$

where τ is the temperature parameter. Then the corresponding loss function is defined as:

$$L_{CE}(\boldsymbol{z}_i, \boldsymbol{a}_i) = -\log p(y_i \mid \boldsymbol{x}_i). \tag{8}$$

For unlabeled data, we adopt the standard consistency regularization paradigm with confidence-based filtering to exclude OOD samples from pseudo-labeling. Given an unlabeled sample $\boldsymbol{x}_i \in \boldsymbol{X}_{\text{unlabeled}}$, we apply weak augmentation to obtain $\boldsymbol{x}_i^w$ and strong augmentation to obtain $\boldsymbol{x}_i^s$. We generate a pseudo-label $\hat{y}_i = \operatorname{argmax}_y p(y \mid \boldsymbol{x}_i^w)$ from the weakly-augmented view and compute its prediction confidence. Only samples with confidence exceeding a threshold ρ are treated as ID samples and retained for consistency regularization:

$$\delta_i = \mathbb{1}[p(\hat{y}_i \mid \boldsymbol{x}_i^w) \geq \rho], \tag{9}$$

where $\delta_i \in \{0, 1\}$ is a binary mask indicating whether the sample should contribute to the loss and $\mathbb{1}(\cdot)$ is the indicator function. The consistency regularization loss is then defined as:

$$L_{CON}(\boldsymbol{a}_i^s) = \delta_i \cdot \mathcal{H}(\hat{y}_i, p(y \mid \boldsymbol{x}_i^s)), \tag{10}$$

where $\mathcal{H}(\cdot, \cdot)$ denotes the cross-entropy loss, $\boldsymbol{a}^s$ denotes the embedding extracted from $\boldsymbol{x}^s$.

Then, we calculate the probability based on the poisoned textual embedding $\boldsymbol{z}_i^p$:

$$p^p(y_i \mid \boldsymbol{x}_i) = \frac{\exp\left(\langle \boldsymbol{z}_i, \boldsymbol{a}_i \rangle / \tau\right)}{\exp\left(\langle \boldsymbol{z}_i, \boldsymbol{a}_i \rangle / \tau\right) + \exp\left(\langle \boldsymbol{z}_i^p, \boldsymbol{a}_i \rangle / \tau\right)}. \tag{11}$$

We employ a contrastive loss to encourage image features to align with the correctly enriched embedding $\boldsymbol{z}_i$ while being repelled from the poisoned embedding $\boldsymbol{z}_i^p$:

$$L_{PC}(\boldsymbol{z}_i^p, \boldsymbol{z}_i) = -\log p^p(y_i \mid \boldsymbol{x}_i). \tag{12}$$

By enforcing contrastive alignment, this mechanism encourages the model to focus on distinctive class-specific cues rather than coarse semantics shared across similar classes. Consequently, near-OOD samples that exhibit only partial or superficial similarities to ID classes are naturally assigned lower confidence scores and filtered out by the confidence-based mechanism in Eq. (9), thereby preventing open-set noise from contaminating the training process.

Training Pipeline: The alignment module serves to bridge the modality gap between patch-level visual tokens and textual embeddings, facilitating accurate memory bank updates. To maintain stable semantic correspondence, we optimize the alignment module using the base textual embeddings $\boldsymbol{b}_i$. The corresponding loss function is defined as:

$$\mathcal{L}_{Memory} = \frac{1}{N} \sum_{i=1}^{N} L_{CE}(\boldsymbol{b}_i, \bar{\boldsymbol{a}}_i) + \frac{\lambda}{M} \sum_{i=1}^{M} L_{CON}(\bar{\boldsymbol{a}}_i^s), \tag{13}$$

where λ is a weighting coefficient.

At the same time, based on the memory bank, we optimize the parameters of the fusion module. The corresponding objective function is defined as:

$$\mathcal{L}_{Total} = \frac{1}{N} \sum_{i=1}^{N} \left[L_{CE}(\boldsymbol{z}_i, \boldsymbol{a}_i) + \nu L_{PC}(\boldsymbol{z}_i^p, \boldsymbol{z}_i) \right] + \frac{\lambda}{M} \sum_{i=1}^{M} L_{CON}(\boldsymbol{a}_i^s), \tag{14}$$

where ν is a weighting coefficient.

Algorithm (1) summarizes the entire training process of our proposed method.

Algorithm 1 Algorithm for our approach.

Input: Labeled set $\boldsymbol{X}_{\text{labeled}}$, unlabeled set $\boldsymbol{X}_{\text{unlabeled}}$, encoders $\phi^v(\cdot)$, $\phi^t(\cdot)$;
Output: Memory bank $\boldsymbol{V}$ and model parameters $\theta = \{\theta_g, \theta_h\}$;
INIT: Initialize memory bank $\boldsymbol{V}$, max iterations T.
for $t = 1$ to T **do**
  for each labeled sample $(\boldsymbol{x}_i, y_i)$ **do**
    Extract $\{\boldsymbol{a}_i^j\}_{j=1}^{n^2}$ and $\boldsymbol{b}_i$ based on $\phi^v(\cdot)$, $\phi^t(\cdot)$.
    Calculate similarity of $\{\boldsymbol{a}_i^j\}_{j=1}^{n^2}$ and $\boldsymbol{b}_i$.
    Calculate $\bar{\boldsymbol{a}}_i$ based on Eq. (3).
    Update memory $\boldsymbol{v}_{y_i}$ with $\bar{\boldsymbol{a}}_i$.
    Calculate $\boldsymbol{z}_i$ based on Eq. (4).
    Calculate $\boldsymbol{z}_i^p$ based on Eq. (6).
    Calculate $L_{CE}(\boldsymbol{z}_i, \boldsymbol{a}_i)$ based on Eq. (8).
    Calculate $L_{PC}(\boldsymbol{z}_i^p, \boldsymbol{z}_i)$ based on Eq. (12).
  end for
  for each unlabeled sample $\boldsymbol{x}_i$ **do**
    Generate $\boldsymbol{x}_i^w$, $\boldsymbol{x}_i^s$.
    Calculate $L_{CON}(\boldsymbol{a}_i^s)$ based on Eq. (10).
  end for
  Update θ_g based on Eq. (13).
  Update θ_h based on Eq. (14).
end for
return Memory bank $\boldsymbol{V}$ and model parameters θ;

4 Experiments

4.1 Experimental Setting

Datasets. We evaluate the proposed approach on five widely-used fine-grained benchmark datasets: CUB-200-2011 [27], Stanford Dogs [13], Stanford Cars [14], FGVC-Aircraft [21] and Semi-Aves [26]. We follow the data split strategy from literature [16]. Specifically, we designate the odd classes as in-distribution classes and the even classes as out-of-distribution classes to minimize domain mismatch between ID and OOD categories. For each ID class, we randomly select 5 or 20 samples from the training set as labeled data, using the remaining training samples as unlabeled data. For evaluation, we apply the same ID/OOD class partition to the test set, and evaluate on all test samples from both ID and OOD classes.

Evaluation Metrics. Following previous works [18, 16], we report two key metrics: (1) classification accuracy on ID classes, which measures the model's ability to correctly classify known categories; (2) open-set classification balanced accuracy, which evaluates ID recognition and OOD detection in a unified metric. Specifically, we treat all unseen classes as a single extra category (i.e., the $(C+1)$-th class), and report the macro-averaged recall over all $C+1$ classes. Additionally, we evaluate OOD detection performance using the Area Under the Receiver Operating Characteristic curve (AUROC) and the false positive rate at a true positive rate of 95% (FPR95) [19]. Higher AUROC and lower FPR95 indicate better OOD detection performance.

Baselines. To illustrate the effectiveness of our method on fine-grained OSSL, we compare it with representative vision-language baselines and recent CLIP-driven OSSL approaches. Specifically, we adopt CLIP [23] as the primary baseline and report its zero-shot performance without fine-tuning. In addition to the CLIP baseline, we consider representative prompt-learning baselines such as CoOp [40], PLOT [3], MaPLe [12] and TextRefiner [31]. We also include LoCoOp [22], a prompt-learning method originally proposed for few-shot OOD detection. Additionally, we evaluate parameter-efficient CLIP adaptation methods, including CLIP-Adapter [4] and CLIP-LoRA [38]. For CLIP-based OSSL baselines, we compare against CFSG-CLIP [16], a recent CLIP-driven framework for fine-grained OSSL.

Implementation Details. We adopt a pretrained ViT-B/16 CLIP model as the backbone. We implement the alignment module as an image adapter followed by a two-layer MLP, and the fusion

Table 1: Classification accuracy on ID classes (%) for CLIP-based methods on four fine-grained benchmark datasets with varying labeled set sizes under the OSSL setting.

Method	Stanford Dogs		Stanford Cars		CUB-200-2011		FGVC Aircraft	
	5	20	5	20	5	20	5	20
CLIP	79.25±0.00	79.25±0.00	75.97±0.00	75.97±0.00	66.00±0.00	66.00±0.00	31.37±0.00	31.37±0.00
CLIP-LORA	83.81±0.37	84.31±0.27	82.71±0.56	82.45±1.30	70.95±0.85	73.40±0.75	40.89±1.73	42.37±0.71
CLIP-Adapter	82.91±0.25	86.02±0.27	84.31±0.02	87.13±0.28	80.03±0.29	84.77±1.02	47.77±0.90	55.79±0.73
CoOp	83.01±0.26	85.68±0.37	85.45±0.31	87.64±0.46	80.10±0.29	85.40±0.37	45.39±0.96	55.43±0.30
LoCoOp	83.08±0.25	86.26±0.11	84.10±0.72	87.83±0.66	79.27±0.45	85.63±0.54	45.53±1.36	54.67±1.59
PLOT	84.46±0.07	87.11±0.09	86.28±0.30	88.59±0.45	81.43±0.66	87.20±0.14	49.59±0.37	58.25±0.93
TextRefiner	85.05±0.20	87.38±0.23	87.18±0.29	89.46±0.36	82.35±0.38	88.05±0.28	50.94±0.54	61.80±0.88
MaPLe	85.64±0.15	87.64±0.20	88.16±0.25	90.34±0.25	83.30±0.33	88.77±0.21	52.43±0.47	64.33±1.21
CFSG-CLIP	85.48±0.21	89.42±0.16	90.38±0.09	93.08±0.08	84.73±0.17	91.75±0.24	61.09±0.27	73.56±0.58
Ours	**86.51**±0.18	**90.24**±0.14	**91.19**±0.12	**93.68**±0.43	**88.09**±0.15	**92.59**±0.19	**61.91**±0.32	**73.90**±0.53

Table 2: Open-set classification balanced accuracy (%) for CLIP-based methods on four fine-grained benchmark datasets with varying labeled set sizes under the OSSL setting.

Method	Stanford Dogs		Stanford Cars		CUB-200-2011		FGVC Aircraft	
	5	20	5	20	5	20	5	20
CLIP	77.17±0.00	77.17±0.00	75.70±0.00	75.70±0.00	64.10±0.00	64.10±0.00	31.08±0.00	31.08±0.00
CLIP-LORA	82.34±0.57	82.67±0.36	82.10±0.56	81.03±1.24	70.52±1.34	72.20±0.45	40.10±1.69	41.57±0.69
CLIP-Adapter	81.63±0.14	84.36±0.25	83.65±0.04	86.50±0.25	81.36±0.60	85.20±0.95	46.87±0.88	53.52±1.35
CoOp	81.65±0.20	84.25±0.41	84.63±0.36	86.84±0.42	81.04±0.06	84.92±0.51	44.55±0.94	54.35±0.30
LoCoOp	81.78±0.23	84.68±0.20	83.25±0.76	87.17±0.64	80.45±0.86	85.46±0.32	44.67±1.35	53.60±1.57
PLOT	82.95±0.07	85.54±0.11	85.58±0.32	87.83±0.34	83.32±0.31	87.16±0.17	48.68±0.37	57.13±0.92
TextRefiner	83.52±0.18	85.78±0.24	86.42±0.28	88.65±0.31	84.03±0.41	87.92±0.35	50.20±0.52	60.45±0.84
MaPLe	84.09±0.19	86.02±0.25	87.43±0.27	89.48±0.25	84.72±0.53	88.66±0.60	51.79±0.43	63.12±1.19
CFSG-CLIP	84.02±0.15	87.77±0.19	89.65±0.05	92.34±0.10	86.46±0.25	90.92±0.24	59.92±0.26	72.13±0.57
Ours	**84.89**±0.17	**88.44**±0.22	**90.34**±0.08	**92.62**±0.13	**86.92**±0.31	**91.79**±0.28	**60.70**±0.33	**73.30**±0.49

module as a single-layer network. For the base prompt, we set the learnable textual prompt length to 16, following the setting in CoOp [40]. All models are trained for 51,200 iterations with a batch size of 32 and a learning rate of 0.002. The consistency regularization loss weight λ is set to 2, and the contrastive loss weight ν is set to 1. Following [25], we use a confidence threshold $\rho = 0.95$ to filter reliable pseudo labels for unlabeled samples. The momentum coefficient μ is set to 0.7, the weighting coefficient α is set to 0.3, and the warmup iterations are set to 10,000. The temperature parameter τ is set to 0.5. We set $K = 5$ for patch-level token selection. All results are presented as the mean with standard deviation over three runs using different random seeds.

4.2 Comparison with State-of-the-Art Methods

Table 1 reports the ID classification accuracy of CLIP-based methods on four fine-grained benchmarks with varying labeled set sizes. Our approach consistently achieves the best results across all datasets and label regimes. Notably, on CUB-200-2011 with only 5 labeled samples per class, we achieve 88.09% accuracy, surpassing the strongest prior approach CFSG-CLIP (84.73%) by 3.36%. We observe similarly consistent gains on other datasets, e.g., our approach improves over CFSG-CLIP by 1.03% on Stanford Dogs (5 samples: 86.51% vs. 85.48%) and achieves the best performance on FGVC Aircraft under both 5 samples (61.91%) and 20 samples (73.90%) settings. These results validate the effectiveness of our memory-driven contrastive embedding enhancement approach. By aggregating fine-grained visual features into a class-wise memory bank and refining textual embeddings with these discriminative cues, our approach achieves superior performance compared to methods relying solely on coarse-grained class-level semantics.

Table 2 summarizes the open-set classification balanced accuracy on four fine-grained benchmarks under the same OSSL setting. Our approach achieves the best balanced accuracy across all datasets and label regimes, demonstrating superior open-set recognition capability under limited supervision. On FGVC Aircraft with 20 labeled samples per class, our approach reaches 73.30% balanced accuracy, outperforming CFSG-CLIP (72.13%) by 1.17%. The gains persist in the 5 samples setting, where our approach achieves 60.70% balanced accuracy compared to 59.92% for CFSG-CLIP. These improvements demonstrate the effectiveness of the approach in strengthening ID-OOD discrimination.

Table 3: Classification accuracy (%) on Semi-Aves under two unlabeled-data settings: ID samples only (U_{in}), and a mixture of ID and OOD samples ($U_{in} + U_{out}$).

Method	U_{in}	$U_{in} + U_{out}$
CLIP	$10.05_{\pm 0.00}$	$10.05_{\pm 0.00}$
CLIP-Adapter	$50.16_{\pm 1.55}$	$50.10_{\pm 0.44}$
CoOp	$47.67_{\pm 0.97}$	$47.93_{\pm 0.76}$
LoCoOp	$48.04_{\pm 0.96}$	$47.30_{\pm 1.30}$
PLOT	$53.68_{\pm 0.35}$	$54.72_{\pm 0.96}$
MaPLe	$60.39_{\pm 0.10}$	$59.68_{\pm 0.64}$
CFSG-CLIP	$65.63_{\pm 0.27}$	$63.69_{\pm 0.74}$
Ours	$\mathbf{67.21}_{\pm \mathbf{0.42}}$	$\mathbf{65.27}_{\pm \mathbf{0.68}}$

Table 4: AUROC and FPR95 (%) on four fine-grained datasets.

Dataset	Labels	AUROC ↑		FPR95 ↓	
		CFSG	Ours	CFSG	Ours
Dogs	5	67.35	**77.48**	88.33	**79.68**
	20	69.29	**77.65**	86.87	**78.83**
Cars	5	63.01	**79.84**	89.67	**76.51**
	20	64.29	**82.35**	88.13	**71.31**
CUB	5	65.78	**76.45**	87.96	**81.47**
	20	71.84	**78.95**	82.17	**77.28**
Aircraft	5	50.16	**53.39**	95.26	**94.57**
	20	54.29	**62.49**	94.98	**90.77**

By contrasting enriched textual embeddings against poisoned ones, our model learns distinctive class-specific features that enable reliable separation between ID and near-OOD samples.

We further evaluate our method on Semi-Aves under two unlabeled-data settings: ID samples only (U_{in}) and a mixture of ID and OOD samples ($U_{in} + U_{out}$). As shown in Table 3, our method achieves the best classification accuracy in both settings, reaching 67.21% and 65.27%, respectively.

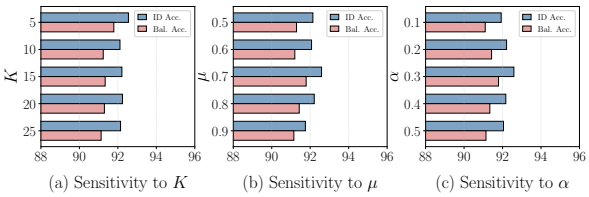
To directly assess OOD detection performance, we report two standard OOD detection metrics, AUROC and FPR95, in Table 4. We compare our method against CFSG-CLIP, the strongest baseline for fine-grained OSSL. Dogs, Cars, CUB, and Aircraft denote Stanford Dogs, Stanford Cars, CUB-200-2011, and FGVC-Aircraft, respectively. Our method consistently achieves higher AUROC and lower FPR95 across all four datasets under both the 5-label and 20-label settings.

4.3 Sensitivity to Hyper-Parameters

We analyze the sensitivity to key hyper-parameters on the CUB-200-2011 dataset with 20 labeled samples per class. We report the classification accuracy on ID classes (ID Acc.) and the open-set classification balanced accuracy (Bal. Acc.) in Figure 3.

Sensitivity to K. As shown in Figure 3 (a), using $K = 5$ achieves the best performance. The performance remains relatively stable across different K values (K from 5 to 25), demonstrating the robustness of our patch-level token selection mechanism.

Sensitivity to μ. Figure 3 (b) shows that our method is robust to the momentum coefficient μ across a wide range of values. Performance remains stable at $\mu \in [0.5, 0.8]$ with ID accuracy consistently above 92%, demonstrating that the visual memory bank updates are not sensitive to the momentum coefficient μ.



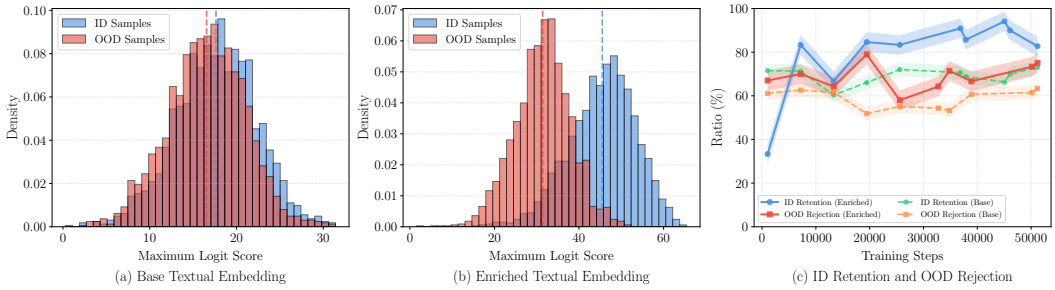
(a) Sensitivity to K (b) Sensitivity to μ (c) Sensitivity to α

Figure 3: Sensitivity analysis on key hyper-parameters. We report results under different settings of K, μ, and α.

Sensitivity to α. Figure 3 (c) demonstrates robustness to the weighting coefficient α. Our approach maintains strong performance across $\alpha \in [0.2, 0.5]$ with ID accuracy above 92%, indicating that the cross-modal injection strategy is not sensitive to the weighting coefficient α.

4.4 Ablation Study

To validate the effectiveness of each component in our approach, we conduct comprehensive ablation studies on the CUB-200-2011 dataset with 20 labeled samples per class. We report the classification accuracy on ID classes (ID Acc.) and the open-set classification balanced accuracy (Bal. Acc.) in Table 5. AM and MU refer to the alignment module and the memory update strategy, respectively.



(a) Base Textual Embedding (b) Enriched Textual Embedding (c) ID Retention and OOD Rejection

Figure 4: ID-OOD separation on CUB-200-2011. (a) Distribution of maximum prediction logits using base textual embeddings. (b) Distribution of maximum prediction logits using enriched textual embeddings. (c) Training dynamics of ID retention and OOD rejection ratios.

Effect of Loss Functions. We progressively incorporate our proposed loss functions to validate their effectiveness. As shown in Rows 1-3 of Table 5, each component consistently improves the final performance. Incorporating the consistency regularization loss L_{CON} increases ID accuracy from 87.80% to 91.27% and balanced accuracy from 86.85% to 90.30%. Further incorporating the contrastive loss L_{PC} yields additional gains, reaching 92.59% ID accuracy and 91.79% balanced accuracy, suggesting that explicitly contrasting semantically consistent versus inconsistent pairs strengthens ID separability and OOD detection.

Table 5: Ablation study on CUB-200-2011 with 20 labeled samples per class.

L_{CE}	L_{CON}	L_{PC}	AM	MU	ID Acc.	Bal. Acc.
✓			✓	✓	87.80	86.85
✓	✓		✓	✓	91.27	90.30
✓	✓	✓	✓	✓	**92.59**	**91.79**
✓				✓	86.64	84.97
✓	✓			✓	88.82	87.42
✓	✓	✓		✓	90.93	90.15
✓			✓		87.35	86.45
✓	✓		✓		89.53	88.55
✓	✓	✓	✓		91.69	90.62

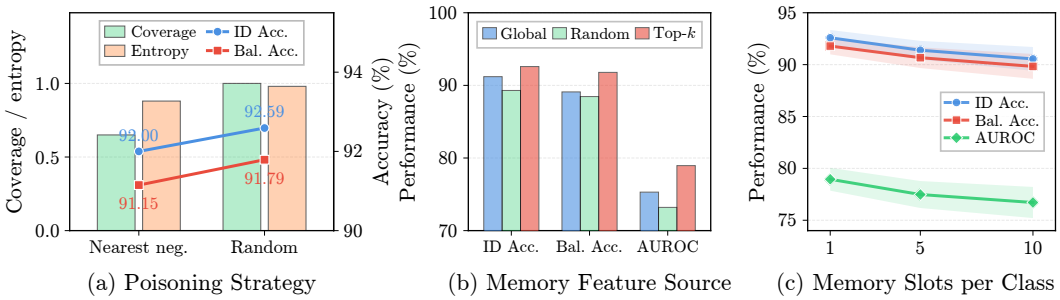
Effect of the Alignment Module. Rows 4-6 of Table 5 examine the role of the alignment module $g(\cdot)$ in bridging the modality gap between patch-level visual tokens and textual embeddings. Without the alignment module, the model achieves 86.64% ID accuracy and 84.97% balanced accuracy using only L_{CE}. Adding L_{CON} improves the performance to 88.82% ID accuracy and 87.42% balanced accuracy, while further incorporating L_{PC} yields 90.93% and 90.15%, respectively. Compared with the full model (Row 3), the remaining gap indicates that the alignment module is critical for effective image-text alignment, bringing a gain of approximately 1.66% in ID accuracy and 1.64% in balanced accuracy. Comparing Rows 1-3 with Rows 4-6 further shows that the alignment module consistently improves both metrics under all three combinations of loss terms.

Effect of Memory Update Strategy. Rows 7-9 of Table 5 evaluate the contribution of memory update strategy. When the memory bank is updated only with labeled samples, the model achieves 91.69% ID accuracy and 90.62% balanced accuracy. Compared with the full model that additionally incorporates high-confidence unlabeled samples for memory updates (Row 3), these results indicate that enriching the memory bank with diverse unlabeled data enables more comprehensive feature aggregation, leading to a gain of approximately 1.2% in balanced accuracy.

4.5 Further Analysis

To better understand how the memory bank improves the distinction between ID and OOD samples, we visualize two aspects in Figure 4: (1) the distribution of maximum prediction logits with base textual embeddings $\{\boldsymbol{b}_i\}$ versus enriched textual embeddings $\{\boldsymbol{z}_i\}$ on the test set, and (2) the training dynamics of ID retention and OOD rejection ratios on the unlabeled dataset. Stronger separation between ID and OOD distributions indicates more effective OOD detection.

As shown in Figure 4 (a), using base textual embeddings results in substantial overlap between ID and OOD distributions, indicating that generic text templates lack sufficient discriminative information to separate the two. In contrast, Figure 4 (b) demonstrates that using enriched textual embeddings



(a) Poisoning Strategy (b) Memory Feature Source (c) Memory Slots per Class

Figure 5: Analysis of poisoning strategies and visual memory bank design on CUB-200-2011 under the 20-label setting. (a) Comparison of random and nearest-negative sampling for constructing poisoned textual embeddings. (b) Comparison of global image features, random patches, and top-k patches for memory construction. (c) Effect of the number of memory slots per class.

achieves much clearer separation, with distinctly separated distributions for ID and OOD samples. Furthermore, Figure 4 (c) reveals that during training, using enriched textual embeddings maintains higher ID retention and OOD rejection ratios than using base embeddings, demonstrating that the memory bank effectively injects discriminative visual information into textual representations.

We further compare different poisoning strategies and visual memory bank designs in Figure 5. Figure 5(a) compares random and nearest-negative sampling for constructing poisoned textual embeddings. Random sampling achieves higher negative-class coverage and sampling entropy, exposing the model to more diverse visual cues and providing richer contrastive signals during training. The higher ID and balanced accuracy further support the effectiveness of diverse negative sampling. Figure 5(b) compares global image features, random patches, and top-K patches for memory construction. Top-K patches perform best across all three metrics, supporting the use of patch-text similarity as a class-relevance prior for memory construction. Figure 5(c) evaluates the effect of the number of memory slots per class. A single slot consistently outperforms five or ten slots, supporting our single-slot memory design.

5 Conclusion

We propose a memory-driven contrastive embedding enhancement approach for fine-grained OSSL, targeting the coarse class-wise semantics that hinder fine-grained ID classification and OOD detection. We design a class-level memory bank that maintains discriminative visual cues for each category. Guided by the memory bank, we inject visual information into textual embeddings through cross-modal contrastive learning, thereby enriching class semantics with fine-grained attributes and improving both ID classification and OOD rejection under open-set noise. Extensive experiments on multiple fine-grained benchmarks demonstrate that our method consistently outperforms strong baselines and achieves state-of-the-art performance.

Limitation. Our visual memory bank is particularly effective when discriminative cues are concentrated in local regions, such as beak shapes or grille designs, which can be captured by top-K patch selection. However, its benefits may provide smaller gains when inter-class differences primarily involve global shapes, such as body proportions or object silhouettes. Incorporating complementary global structural cues into the memory mechanism remains a direction for future work.

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A An Alternative Use of the Visual Memory Bank: Pseudo-OOD Synthesis

In this section, we explore an alternative way to leverage the visual memory bank for enhancing ID-OOD discrimination. Beyond the contrastive embedding enhancement presented in the main paper, the memory bank can also be utilized to synthesize pseudo-OOD samples.

A.1 Pseudo-OOD Construction

Given a labeled sample $(\boldsymbol{x}_i, y_i)$ with its corresponding visual embedding $\boldsymbol{a}_i = \phi^v(\boldsymbol{x}_i)$ and the class-wise memory bank $\boldsymbol{V} = \{\boldsymbol{v}_1, \cdots, \boldsymbol{v}_C\}$, we construct a pseudo-OOD sample by mixing $\boldsymbol{a}_i$ with a memory vector from a different class. Specifically, we first randomly select a class $c \neq y_i$, and then synthesize a pseudo-OOD feature as:

$$\tilde{\boldsymbol{a}}_i = \beta \cdot \boldsymbol{a}_i + (1 - \beta) \cdot \boldsymbol{v}_c, \tag{15}$$

where $\beta \in [0, 1]$ is a mixing coefficient that controls the degree of semantic conflict. $\tilde{\boldsymbol{a}}_i$ intentionally contains conflicting visual semantics from two distinct classes. This semantic conflict simulates a key property of near-OOD samples: they often exhibit partial similarities to multiple ID classes while not belonging to any of them, making them difficult to filter out as they may receive high confidence from the model.

A.2 Uniform Distribution Loss

To encourage the model to assign low and uniform confidence scores to pseudo-OOD samples, we employ a uniform distribution loss. Specifically, given the synthesized feature $\tilde{\boldsymbol{a}}_i$ and the enriched textual embeddings $\{\boldsymbol{z}_1, \cdots, \boldsymbol{z}_C\}$, we compute the predicted probability distribution:

$$p(j \mid \tilde{\boldsymbol{a}}_i) = \frac{\exp(\langle \tilde{\boldsymbol{a}}_i, \boldsymbol{z}_j \rangle / \tau)}{\sum_{k=1}^{C} \exp(\langle \tilde{\boldsymbol{a}}_i, \boldsymbol{z}_k \rangle / \tau)}, \quad j \in \{1, \ldots, C\}, \tag{16}$$

where τ is the temperature parameter.

The uniform distribution loss is then defined as the cross-entropy between a uniform target distribution and the predicted distribution:

$$L_{UNI}(\tilde{\boldsymbol{a}}_i) = -\sum_{j=1}^{C} \frac{1}{C} \log p(j \mid \tilde{\boldsymbol{a}}_i). \tag{17}$$

By minimizing L_{UNI}, the model learns to avoid making confident predictions on semantically conflicting samples, thereby naturally filtering out near-OOD samples during training.

A.3 Training Integration

The uniform distribution loss is incorporated into the objective for updating the fusion module with a weighting coefficient λ_{ood}:

$$\mathcal{L}_{Total} = \frac{1}{N} \sum_{i=1}^{N} \Big[L_{CE}(\boldsymbol{z}_i, \boldsymbol{a}_i) + \lambda_{\text{ood}} L_{UNI}(\tilde{\boldsymbol{a}}_i) \Big] + \frac{\lambda}{M} \sum_{i=1}^{M} L_{CON}(\boldsymbol{a}_i^s). \tag{18}$$

where L_{CE} and L_{CON} are defined in the main paper, and the alignment module is optimized using the same objective as the main paper.

We validate this pseudo-OOD synthesis module in Appendix C.2. Note that this module is used only for additional analysis and is not used in the main experiments.

B Notation and Datasets

In this section, we summarize the key notation used throughout the paper and provide additional details on the datasets used in our experiments.

Table I: Notation Definition.

Symbol	Description
$\boldsymbol{X}_{\text{labeled}} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^{N}$	Labeled training set with N samples
$\boldsymbol{X}_{\text{unlabeled}} = \{\boldsymbol{x}_i\}_{i=1}^{M}$	Unlabeled set with M samples
$\mathcal{D}_{\text{labeled}}$	Data distributions for labeled sets
$\mathcal{D}_{\text{unlabeled}}$	Data distributions for unlabeled sets
$\boldsymbol{x}_i$	Input image sample
C	the number of ID classes
$y_i \in \{1, \dots, C\}$	ID class label
$\phi^v(\cdot)$	Image encoder
$\phi^t(\cdot)$	Text encoder
$\boldsymbol{t}_i$	Label-based textual prompt for class y_i
$\boldsymbol{a}_i$	Global image embedding of x_i
$\boldsymbol{b}_i$	Base textual embedding of prompt t_i
n^2	Number of non-overlapping image patches
$\boldsymbol{x}_i^j$	The j-th patch of image x_i
$\boldsymbol{a}_i^j$	Patch-level token
K	Number of selected top patches
$g(\cdot;\theta_g)$	Alignment module with parameters θ_g
$h(\cdot;\theta_h)$	Fusion module with parameters θ_h
$\boldsymbol{V}$	Visual memory bank
$\boldsymbol{v}_{y_i}$	Memory vector of class y_i
μ	Momentum coefficient
$[\cdot;\cdot]$	Vector concatenation operator
α	Weighting coefficient
$\boldsymbol{z}_i$	Enriched textual embedding
ξ_i	Randomly sampled memory vector
$\boldsymbol{z}_i^p$	Poisoned textual embedding
τ	Temperature parameter
ρ	Confidence threshold for filtering unlabeled samples
$\mathbb{1}(\cdot)$	Indicator function
$H(\cdot,\cdot)$	Cross-entropy operator
L_{CE}	Cross-entropy loss
L_{CON}	Consistency regularization loss
L_{PC}	contrastive loss
λ	Weighting coefficient for L_{CON}
ν	Weighting coefficient for L_{PC}

B.1 Notation

Table I summarizes the key notation used throughout the paper.

B.2 Datasets

Stanford Dogs. Stanford Dogs [13] is a fine-grained dog categories recognition benchmark containing 20,580 images from 120 dog breeds. Following the standard split, it provides 12,000 training images and 8,580 test images, and includes breed labels and bounding-box annotations for the training set. We use the official train/test split for all experiments.

Stanford Cars. Stanford Cars [14] consists of 16,185 images from 196 car categories. The dataset is officially split into 8,144 training images and 8,041 testing images, with an approximately balanced per-class partition. We use the official train/test split for all experiments.

CUB-200-2011. CUB-200-2011 [27] is a widely used fine-grained bird classification dataset with 200 bird categories and 11,788 images in total. The official split contains 5,994 training images and 5,794 testing images. In addition to category labels, it provides rich instance-level annotations, including one bounding box per image, 15 part locations, and 312 binary attributes. We use the official train/test split for all experiments.

FGVC-Aircraft. FGVC-Aircraft [21] is a fine-grained aircraft recognition benchmark comprising 10,000 images across 100 aircraft categories, with each image annotated by a tight bounding box and a hierarchical aircraft model label. The standard protocol partitions images into train/validation/test subsets with roughly equal size per variant (typically $33/33/34$ images per category). We use the official train/test split for all experiments.

C Additional Experimental Results

C.1 Visualization of Selected Patches

To provide insights into what fine-grained visual cues are captured by our visual memory bank, we visualize the top-K selected patches on random samples from four fine-grained datasets (in practice, we set $K = 5$): CUB-200-2011 [27], Stanford Cars [14], Stanford Dogs [13] and FGVC-Aircraft [21]. Figure I shows that the selected patches tend to focus on discriminative local regions that are characteristic of each category.

For bird species (CUB-200-2011) [27], the model selects patches corresponding to distinctive features such as beak shape, eye rings, and plumage patterns. For car models (Stanford Cars) [14], the selected patches concentrate on headlights, grilles, and body contours. For dog breeds (Stanford Dogs) [13], the model focuses on facial features, ear shapes, and fur textures. For aircraft variants (FGVC-Aircraft) [21], the selected patches primarily attend to the fuselage body, cockpit and window regions. These visualizations indicate that our visual memory bank effectively aggregates the most distinctive visual cues for each category.

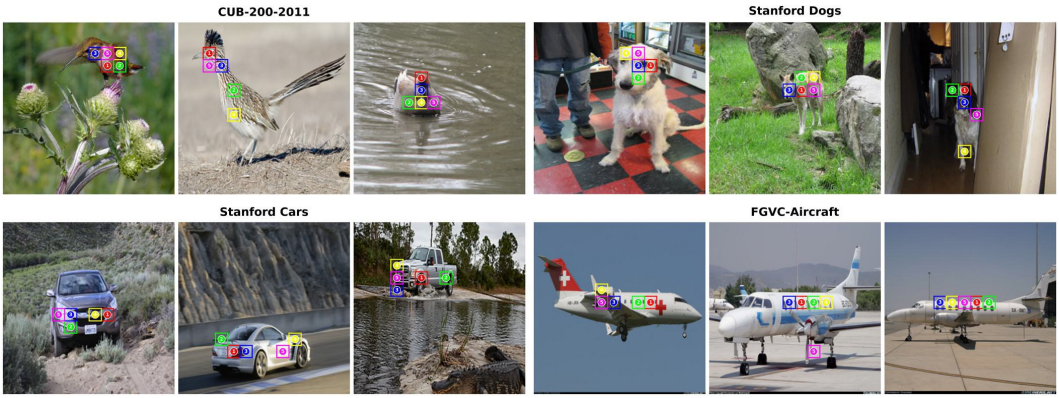


Figure I: Visualization of top-5 selected patches on representative samples. The colored boxes indicate the patches with highest cosine similarity to the textual embedding, numbered by their ranking.

C.2 Impact of Pseudo-OOD Synthesis

To validate the effectiveness of the memory-driven pseudo-OOD synthesis module (Appendix A), we disable the contrastive embedding enhancement component used in the main paper and rely solely on the memory bank for pseudo-OOD synthesis. We then compare models trained with and without this mechanism. We set the mixing coefficient $\beta = 0.5$ and the weighting coefficient $\lambda_{\text{ood}} = 0.1$.

We report the ID classification accuracy (ID Acc.) and the open-set classification balanced accuracy (Bal. Acc.) in Table II. Incorporating pseudo-OOD synthesis brings consistent improvements across all four datasets. Specifically, it increases the ID classification accuracy by roughly 0.24%-1.17% and also improves balanced accuracy by about 0.50%-1.88%. This suggests that semantically conflicting pseudo-OOD training improves both OOD rejection and ID discrimination, yielding more reliable open-set decision boundaries.

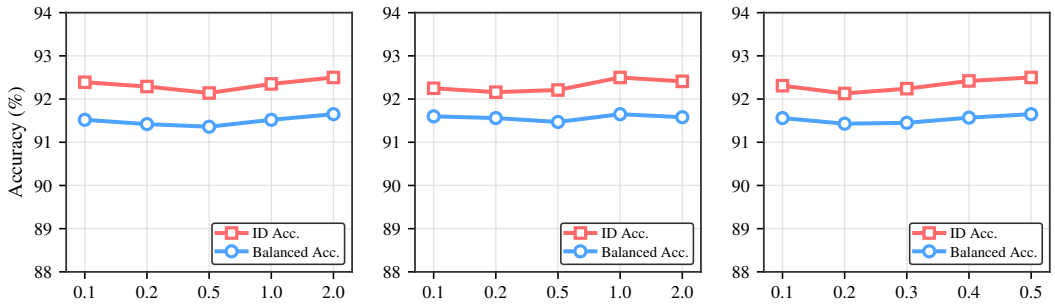
Table II: Effect of memory-driven pseudo-OOD synthesis on four datasets (20 labeled samples).

Dataset	Method	ID Acc. (%)	Bal. Acc. (%)
Stanford Dogs	w/o Pseudo-OOD	89.23	87.55
	w/ Pseudo-OOD	**89.92**	**88.05**
Stanford Cars	w/o Pseudo-OOD	92.33	91.46
	w/ Pseudo-OOD	**93.25**	**92.20**
CUB-200-2011	w/o Pseudo-OOD	91.27	90.30
	w/ Pseudo-OOD	**91.51**	**91.28**
FGVC-Aircraft	w/o Pseudo-OOD	72.28	70.92
	w/ Pseudo-OOD	**73.45**	**72.80**

C.3 Sensitivity to Hyper-parameters

We report additional sensitivity analyses for three hyper-parameters: the weighting coefficient λ for consistency regularization loss, ν for contrastive loss, and temperature parameter τ. All experiments are conducted on the CUB-200-2011 dataset under the 20-labeled setting.

As shown in Figure II, our method remains stable over a broad range of hyper-parameters. The results indicate that our approach is not sensitive to hyper-parameters.



(a) Weighting Coefficient λ (b) Weighting Coefficient ν (c) Temperature Parameter τ

Figure II: Sensitivity analysis of hyper-parameters on CUB-200-2011 (20 labeled samples).

C.4 Textual Discriminability Analysis

To understand how the memory bank enhances textual discriminability, we visualize inter-class similarity matrices as heatmaps in Figure III. For simplicity, we randomly sample 6 ID classes from CUB-200-2011 [27].

We compute three types of similarity matrices: (1) visual memory bank feature vectors $\boldsymbol{v}_{y_i}$, (2) base textual embeddings $\boldsymbol{b}_c$, and (3) enriched textual embeddings $\boldsymbol{z}_i$ that incorporate visual cues from the memory bank. We quantify discriminability by computing the average dissimilarity between different classes. Higher values indicate better inter-class separation. We observe that base textual embeddings exhibit poor discriminability (7.2%) with a nearly uniform red heatmap, indicating that generic text templates fail to distinguish fine-grained categories. In contrast, memory bank feature vectors achieve near-perfect discriminability (above 95%) with clear blue off-diagonal regions. After fusion, the enriched textual embeddings substantially improve to 58.6% discriminability, demonstrating that the memory bank effectively injects discriminative visual information into textual representation.

C.5 Computational Complexity

All experiments in the main paper are conducted on a single NVIDIA RTX 4090 GPU. We further compare the computational overhead of our method with the strongest baseline, CFSG-CLIP [16], on CUB-200-2011 [27] under the 20-label setting. As shown in Table III, our method improves the ID

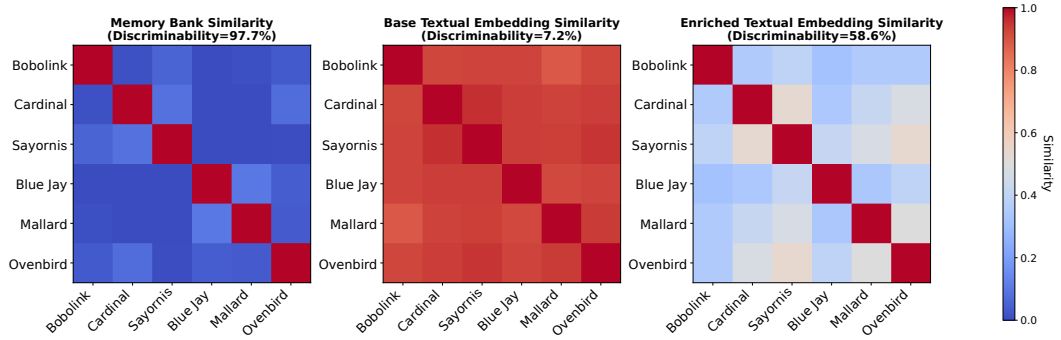


Figure III: Textual discriminability analysis on CUB-200-2011. Left: similarity matrix for visual memory bank feature vectors. Middle: similarity matrix for base textual embeddings. Right: similarity matrix for enriched textual embeddings.

classification accuracy from 91.75% to 92.59% while maintaining competitive computational cost, with comparable training time and lower memory usage than CFSG-CLIP.

Table III: Computational overhead comparison on CUB-200-2011 under the 20-label setting.

Method	ID Acc. (%)	Extra Params (M)	Train Time (h)	Train Mem (GB)	Infer Mem (GB)
CLIP	66.00	0.0	–	–	3.4
CFSG-CLIP	91.75	0.9	11.0	12.3	4.2
Ours	**92.59**	1.2	12.5	7.2	3.8

C.6 Relationship to the CLIP Textual Embedding Space

We further analyze the relationship between enriched textual embeddings and the original CLIP textual embedding space. We measure the cosine similarity between each enriched textual embedding and its corresponding base textual embedding, and use Pearson and Spearman correlations to assess the preservation of inter-class relationships. As shown in Table IV, the Pearson and Spearman correlations suggest that the enriched textual embeddings largely preserve the inter-class semantic structure of the original CLIP text embeddings.

Table IV: Relationship between enriched textual embeddings and the original CLIP text embedding space across four fine-grained datasets.

Dataset	Cosine sim.	Pearson	Spearman
Stanford Dogs	0.938	0.819	0.818
Stanford Cars	0.941	0.787	0.770
CUB-200-2011	0.942	0.801	0.764
FGVC-Aircraft	0.965	0.794	0.784

C.7 Near- and Far-OOD Detection on CIFAR-100

To directly assess performance on near- and far-OOD samples, we construct matched OOD subsets on CIFAR-100 using its superclass hierarchy. Near-OOD classes belong to the same superclasses as the ID classes, whereas far-OOD classes belong to superclasses disjoint from those containing ID classes. As shown in Table V, our method achieves higher AUROC and lower FPR95 than CFSG-CLIP on both subsets. These results demonstrate improved discrimination between ID and challenging near-OOD samples.

Table V: Near- and far-OOD detection on CIFAR-100.

Method	Near-OOD AUROC (%) ↑	Far-OOD AUROC (%) ↑	AUROC gap (pp) ↓	Near-OOD FPR95 (%) ↓	Far-OOD FPR95 (%) ↓
CFSG-CLIP	64.98	82.38	17.40	89.65	76.70
Ours	**71.09**	**84.65**	**13.56**	**87.60**	**68.94**

C.8 Analysis of Cross-Modal Injection Strategies

We further compare our method with a straightforward alternative cross-modal injection strategy. For each input image, this alternative directly concatenates its visual feature with the corresponding textual feature for classification, without using the alignment module, proposed memory bank, or fusion module. Therefore, the visual cues used by this simpler design are entirely derived from a single image. Although this strategy introduces additional visual information, it does not explicitly select the most discriminative patch-level cues or aggregate stable class-level information from multiple samples. As a result, the concatenated textual embedding can be easily dominated by instance-specific noise, background regions, or non-discriminative visual details.

In contrast, our method first selects the most relevant patch-level cues from different images and maps them with the alignment module before averaging them into class-wise memory vectors. These memory vectors are then injected into textual embeddings through the designed fusion module. As shown in Table VI, under the 20-label setting on CUB-200-2011 [27], single-image direct feature concatenation achieves 89.51% ID classification accuracy, while our design improves the result to 92.59%. This comparison indicates that the improvement does not come from simply appending visual features, but from the proposed aggregation and fusion design.

Table VI: Comparison with direct per-image feature concatenation on CUB-200-2011 under the 20-label setting.

Method	ID Acc. (%)	Bal. Acc. (%)
Direct per-image feature concatenation	89.51	88.65
Ours	**92.59**	**91.79**

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